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Study on the Balance of Tension between Ropes of Four-rope Freight Ropeway

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Abstract. Based on the analysis of spatial position of the working-rope for four-rope freight ropeway, a mechanical model of the interaction between the working-rope and running car is proposed, which comprehensively reflects the characteristics of the working-rope system. Based on the calculation method of elastic catenary, the nonlinear equations for analysing the working system of four-rope freight ropeway are established. According to the typical design conditions of four-rope ropeway, the tension calculation of four carrying ropes is completed when the length of carrying rope changes. With the increase of the length deviation between the carrying ropes, the imbalance of the tension between the ropes is obvious. The tension of the carrying rope of the four-rope ropeway is very sensitive to the change of the rope length, and the coupling effect between ropes makes the breaking risk of carrying rope increase sharply. The results provide a reference for the safety factor analysis of the four-rope freight ropeway.

1. Introduction

The transport capacity of the freight ropeway mainly depends on the bearing capacity of the carrying ropes. At present, the heavy freight ropeways mainly use multiple (two or four) small steel wire ropes as the carrying ropes at the same time, so as to meet the transportation of large and heavy goods [1].

Freight ropeway mainly includes working-rope (carrying rope and haulage rope), trestle, running car, tractor and other parts. When there are more than two trestles, it is called multi-span ropeway. The analysis of tension and rope shape of working-rope is the key to ropeway design. In order to solve the nonlinear problems of large deformation of suspension structures, domestic and foreign scholars put forward the finite element method [2], analytic method [3-5] and other calculation methods, and realized the analysis of the tension and sag of flexible suspension ropes. In addition, the dynamic calculation method of suspension rope with moving load [6-8] is also one of the research hotspots, and a series of achievements have been achieved.

At present, the total carrying capacity is still obtained by multiplying the carrying capacity of single rope by the number of ropes in the calculation method of four-rope freight ropeway. The interaction relationship between the carrying ropes and the influence of the haulage rope on the tension of the carrying rope cannot be analysed during the operation of the ropeway.

Therefore, this paper establishes the overall mechanical model of four-rope system, studies the calculation method of multi-rope tension, and analyses the carrying capacity and safety performance of four-rope freight ropeway according to typical conditions.



2. Structure and Parameters of Four-rope Freight Ropeway

2.1. Ropeway Structure

Four-rope freight ropeway is erected in parallel with four carrying ropes to bear the load (trolley and cargo) of the ropeway. The haulage rope is located under the carrying rope, and the running car moves along the carrying rope. The trestle is set up at the beginning, end and middle of the ropeway, and the carrying rope saddle and roller are installed under the trestle beam to support the working-rope. The haulage rope can pull the running car and load through the trestle, as shown in Figure 1.



Figure 1. Four-rope freight ropeway

2.2. Basic Parameters of Ropeway

For the ropeway with $n+1$ trestles, the span is L_i^0 and the height difference is h_i^0 ($i = 1, 2, \dots, n$), as shown in Figure 2.

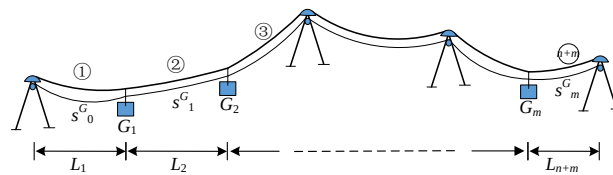


Figure 2. Load and haulage rope parameters

The total number of loads in the ropeway is m and the gravity is G_i ($i = 1, 2, \dots, m$). Set the load to be connected by a haulage rope with a length of s_i^G ($i=1, \dots, m-1$) between the loads. At the same time, the length of the haulage rope section from the start of the ropeway to the first load is s_0^G and the length of the traction section from the last load to the end of the ropeway is s_m^G , as shown in Figure 2.

3. Mechanical Model of Four-rope Freight Ropeway

3.1. Carrying Rope and Running Car

During the operation of the freight ropeway, the carrying rope and the running car are always in contact, as shown in Figure 3.

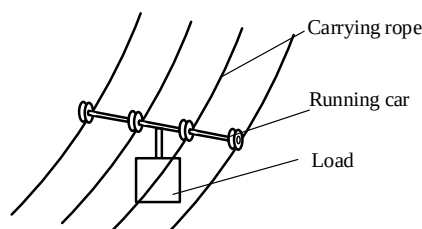


Figure 3. Carrying rope and running car

3.2. Deformation of Working-rope

Each section of carrying rope and haulage rope can be represented by an elastic catenary equation. According to literature [3], the span and height difference of elastic catenary segments subjected only to gravity can be expressed as:

$$L = \frac{H}{q} \left[\frac{qs}{EA_0} + \operatorname{asinh} \frac{V_B}{H} - \operatorname{asinh} \frac{V_A}{H} \right] \quad (1)$$

$$h = \frac{V_B^2 - V_A^2}{2qEA_0} + \frac{H}{q} \left(\sqrt{1 + \frac{V_B^2}{H^2}} - \sqrt{1 + \frac{V_A^2}{H^2}} \right) \quad (2)$$

Where q is the unit gravity of the rope; EA_0 is the product of cross-sectional area and elastic modulus; s is the initial length of the rope section; H is the horizontal tension of the rope section; V_A is the vertical tension of point A; V_B is the vertical tension of point B, it can be expressed as:

$$V_B = V_A + qs \quad (3)$$

In addition, the tangential tension of the rope segment can be expressed as:

$$T = \sqrt{H^2 + V^2} \quad (4)$$

It can be seen that the main parameters such as span, height difference and tangential tension of rope segment can be expressed by the length s , horizontal tension H and vertical tension V_A of point A. In the parameter, the superscript L is the related variable of the carrying rope, and the superscript t is the related variable of haulage rope. The segment number is at the end of the subscript.

4. Structural Equation of Four-rope Freight Ropeway

During the operation of ropeway, some parameters related to the overall structure of carrying rope and haulage rope will not change. The corresponding equations can be established from the following aspects.

4.1. Height Difference and Span

The sum of height difference and span of each rope section in the span is equal to the height difference and span of supports at the ends of each span.

4.2. Span and Height Difference in Each Rope Section

The span of each rope section is the same, but due to the deflection of the running car, the height difference between the carrying rope section and the haulage rope section is not equal, as shown in Figure 4. Therefore, the equation can be established according to the relative positions of the starting and ending points of the carrying rope section and the haulage rope section, and also the deflection angle of the running car.

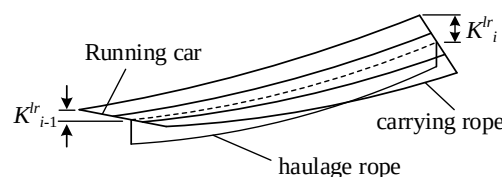


Figure 4. Span and height difference of each section of carrying rope and haulage rope

4.3. Length of Haulage Rope between Loads

The load is connected by the haulage rope, and the load may include one or more haulage rope sections during operation.

4.4. Length of Carrying Rope

The total length of the carrying rope remains unchanged.

4.5. Force Balance of Running Car

The car is balanced under the combined action of carrying rope, haulage rope and load gravity. The force balance equation can be established in the horizontal and vertical directions.

4.6. Tension Balance of Working-rope

The roller of running car only produces lateral pressure on the carrying rope, which makes the carrying rope bend, but has no effect on the axial force. Therefore, the tangential tension of the rope is equal on both sides of the load.

When the haulage rope moves on the roller of the trestle, the tangential tension of the traction cable in front of the support is equal to that in the back of the bracket, ignoring the rolling friction.

5. Four-rope Coupling Tension Analysis Method

Based on the above analysis of multi-rope structure and force balance relationship, the equations for four-rope tension coupling can be established.

The following parameters are taken as unknown variables: Length of each section of carrying rope is s_i^{lr} , the horizontal tension is H_i^{lr} and the vertical tension at point A is V_{Ai}^{lr} ($i=1, \dots, n+m$), where r stands for carrying rope number ($r=1, 2, 3, 4$); Length of each section of the haulage rope (including haulage rope and return rope) is s_i^t , horizontal tension H_i^t , vertical tension at point A is V_{Ai}^t , ($i=1, \dots, n+m$); deflection angle of running car is ω_i ($i=1, \dots, m$).

The number of nonlinear equations obtained is equal to the number of unknown variables, so the equations are closed. It can be expressed as:

$$F(X) = 0 \quad (5)$$

Where $X=[s_1^{lr}, s_2^{lr}, \dots, s_{n+m}^{lr}, H_1^{lr}, H_2^{lr}, \dots, H_{n+m}^{lr}, V_{A1}^{lr}, \dots, V_{An+m}^{lr}, s_1^t, \dots, s_{n+m}^t, H_1^t, \dots, H_{n+m}^t, V_{A1}^t, \dots, V_{An+m}^t]^T$.

The nonlinear equations can be solved by Newton's iteration method.

6. Analysis of Four-rope Coupling Tension

For freight ropeway with multiple carrying rope, the interaction among the carrying rope, haulage rope and running car should be considered in order to accurately calculate the bearing capacity of the working-rope and obtain the safety factor, and put forward the appropriate design requirements for the structural components of freight ropeway.

In this paper, combined with the typical working conditions of the freight ropeway with two span and four-rope, the tension effect caused by the change of the length for the carrying rope is analysed.

The first span is 285m and the height difference is 50m; The second span is 20m and the height difference is 5m. The diameter of the carrying rope is 22mm, the elastic modulus is 110GPa, and the mass per unit length is 1.8kg/m; The diameter of the haulage rope is 18 mm, the elastic modulus is 100 GPa, and the mass per unit length is 0.6 kg/m.

(1) Basic conditions

When the length of the four carrying ropes is 306m and the length of the haulage rope is 308m, the maximum tension of the carrying rope is located at the left side of the load, 150m from the starting point, and the maximum tension of the carrying rope is 113.22kN.

(2) Coupling tension analysis

1) The carrying rope 1, 2 and 3 are 306m, and rope 4 is changed from 306m to 307m

When the length of rope 4 changes, the maximum tension value of each carrying rope and the deflection angle of running car at the maximum tension are shown in Figure 5 and Figure 6.

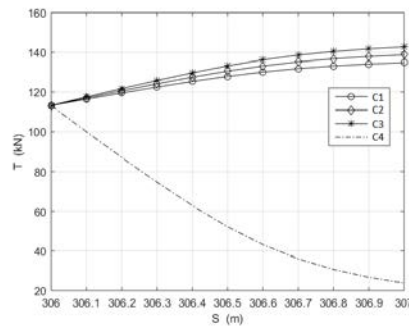


Figure 5. The tension of each cable varies with the length of the rope 4

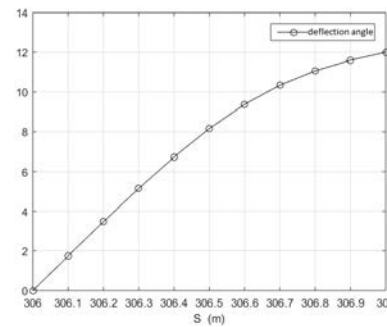


Figure 6. The deflection angle of the trolley varies with the length of the rope 4

It can be seen from Figure 5 that due to the increase of rope 4, the load is mainly borne by rope 1, 2 and 3. The tension of rope 3 increases by 29.49kN (increased by 26.04%), while the tension of rope 4 decreases rapidly.

2) The carrying rope 1, 2 and 3 are 306m, and rope 4 is changed from 306m to 305m

When the length of rope 4 changes, the maximum tension value of each carrying rope and the deflection angle of running car at the maximum tension are shown in Figure 7 and Figure 8. It can be seen that the tension of the carrying rope is very sensitive to the length of the rope.

Due to the shortening of carrying rope 4, the load is mainly borne by rope 4, the tension increases by 137.72kN (increased by 121.63%), and the tension of other carrying ropes decreases rapidly. This kind of working condition is easy to cause the breakage of rope 4, which has a great effect on the safety of ropeway.

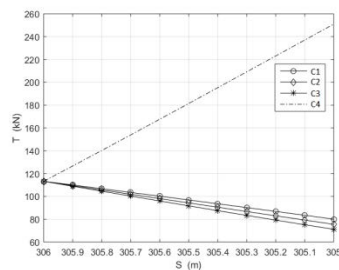


Figure 7. The tension of each cable varies with the length of the rope 4

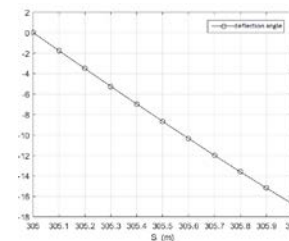


Figure 8. The deflection angle of the trolley varies with the length of the rope 4

3) The carrying rope 1, 2 and 4 are 306m, and rope 3 is changed from 306m to 305m

When the length of rope 3 changes, the maximum tension value of each carrying rope and the deflection angle of running car at the maximum tension are shown in Figure 9 and Figure 10.

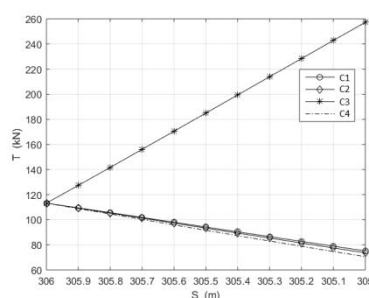


Figure 9. The tension of each cable varies with the length of the rope 3

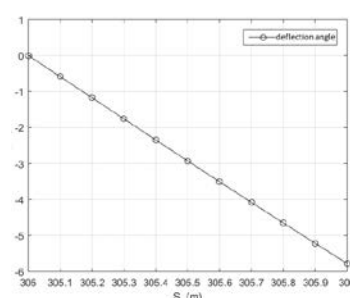


Figure 10. The deflection angle of the trolley varies with the length of the rope 3

Due to the shortening of carrying rope 3, the load is mainly borne by rope 3, the tension of which increases 144.01kN (127.19%) and the tension of other carrying ropes decreases rapidly. This kind of working condition is very easy to cause the breakage of rope 3, and then affect other carrying ropes, which will have a great effect on the safety of ropeway.

7. Summary

The analysis method of working-rope coupling system for four-rope freight ropeway is proposed in this paper, which can realize the calculation of coupling tension between four carrying ropes and grasp the bearing performance of freight ropeway with multiple bearing cables more comprehensively.

According to the typical design conditions of four carrying ropes ropeway, the tension calculation of four carrying ropes is completed when the length of carrying rope changes. Through calculation, it can be seen that the tension of carrying rope of four cable ropeway is very sensitive to the change of rope length. With the increase of the length deviation between the carrying ropes, the imbalance of the tension between the ropes is obvious. The tension change caused by the coupling effect between four ropes makes the breakage risk of carrying ropes increase sharply.

Therefore, in the design of multi-rope heavy freight ropeway, the influence of tension caused by the length error of different carrying ropes should be fully considered, and the reasonable tension safety factor should be adopted to ensure the safety of construction and transportation.

8. Acknowledgement

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9. References

- [1] MIAO Qian, BAI Xuesong. Study on transportation technology and equipment in freight cableway system. *Electric power construction*, 2009, 30(12), pp 93-96.
- [2] YANG Menggang, CHEN Zheng-qing. Nonlinear analysis of cable structures using a two-node curved cable element of high precision. *Engineering Mechanics*, 2003, 20(1), pp 42-47.
- [3] ZHANG Zhiguo, JIN Mingjun, ZOU Zhenzhu. Static Solution of Suspension Cables under Tare Load. *China Railway Science*, 2004, 25(3), pp 67-70.
- [4] Chen Changsong, Chen Zhengqing, Yan Donghuang. Accurate iteration method to calculate the initial. *Engineering Mechanics*, 2006, 23(8), pp 62-68
- [5] Qin Jian, Xia Yongjun. The Matrix Iteration Method for Analysis of Suspension Cable Based on Segmental Category Theory. *Chinese Journal of Engineering Design*, 2013, 20(5), pp 404-408.
- [6] Wang Y. M. The transient dynamics of a cable-mass system due to the motion of an attached accelerating mass. *International Journal of Solids and Structures*, 2000, 37, pp 1361—1383.
- [7] Al-Qassab, M., Nair, S., O'Leary. Dynamics of an elastic cable carrying a moving mass particle. *Nonlinear Dynamics*, 2003, 33, pp 11—32.
- [8] Lianhua Wang, Giuseppe Rega. Modelling and transient planar dynamics of suspended cables with moving mass. *International Journal of Solids and Structures*. 2010, 47, pp 2733—2744.